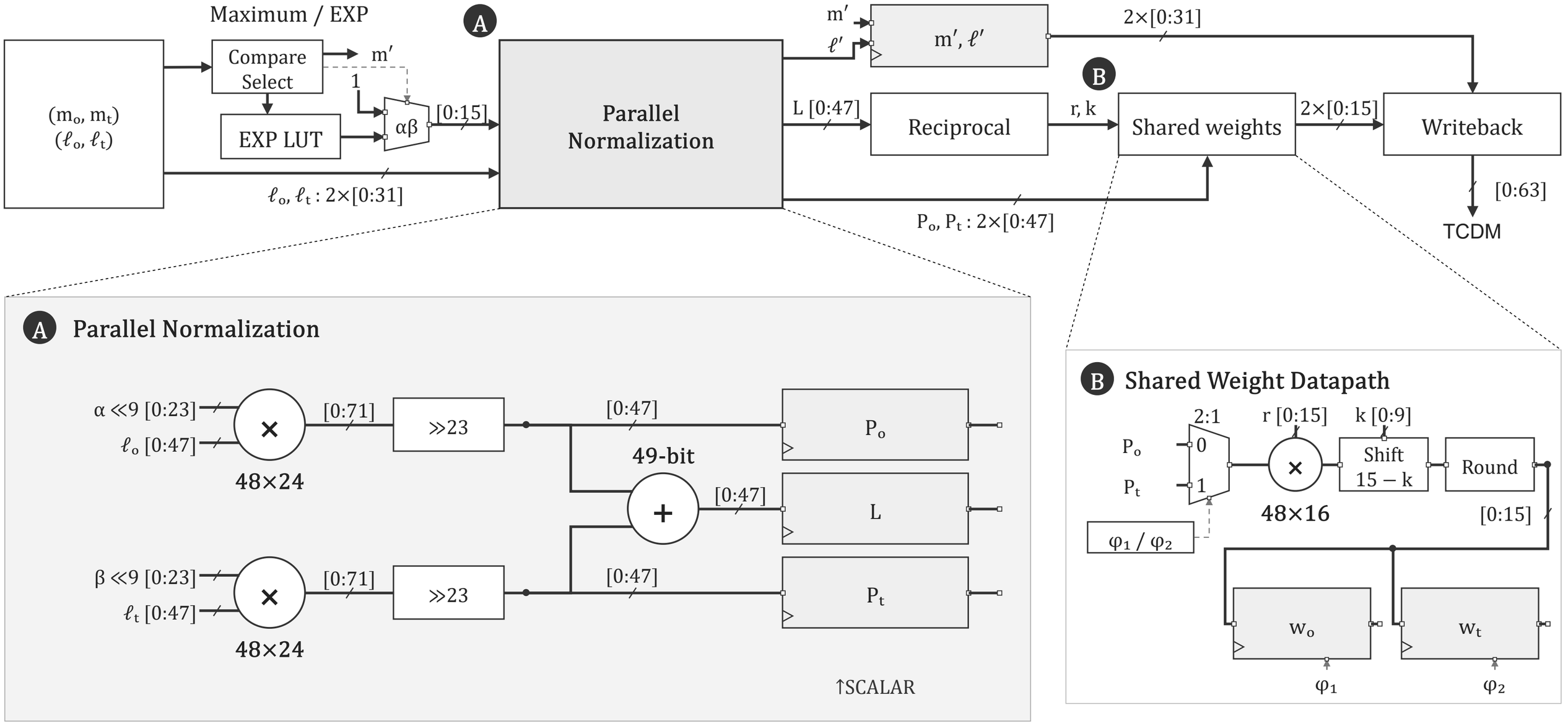

Maximum / EXP
A
(m_o, m_t)
(ℓ_o, ℓ_t)
Compare Select
m′
1
EXP LUT
αβ
[0:15]
ℓ_o, ℓ_t : 2×[0:31]
Parallel Normalization
m′
ℓ′
m′, ℓ′
2×[0:31]
B
L [0:47]
Reciprocal
r, k
Shared weights
2×[0:15]
Writeback
[0:63]
TCDM
P_o, P_t : 2×[0:47]
A Parallel Normalization
α ≪9 [0:23]
ℓ_o [0:47]
48×24
[0:71]
≫23
[0:47]
49-bit
P_o
[0:47]
L
β ≪9 [0:23]
ℓ_t [0:47]
48×24
[0:71]
≫23
[0:47]
P_t
↑SCALAR
B Shared Weight Datapath
2:1
r [0:15]
k [0:9]
P_o
P_t
0
1
Shift 15 − k
Round
48×16
[0:15]
φ1 / φ2
w_o
w_t
φ1
φ2